API Common Usage & Style Guide

API User Document

Version 1.0

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Purpose

This common usage and style guide describes the consistent URL structure, location types, language support and terminology used by the TWC APIs. This document provides the standards for the construction of each API's URL for customer use.

Host Name

api.weather.com

Geography

Worldwide.

Background Technology

This API is a REST-based web service.

Response Format

This TWC API can return either JSON or XML responses.

Icon Codes, Weather Phrases and Images

For the mapping of icon codes, weather phrases and images please refer to the Icon Codes, Weather Phrases and Images document. To access the standard, high-resolution TWC weather icons, please refer to this shared folder.

Data Lifetime - Caching & Expiration

Standard HTTP Cache-Control headers are used to define caching length. The TTL value is provided in the HTTP Header as an absolute time value using the "Expires" parameter, for example: "Expires: Fri, 12 Jul 2013 12:00:00 GMT". The response provides a data element expire_time_gmt. The value in this data element should be used to expire and remove a record from your system.

Worldwide Locations

TWC provides weather content worldwide. The content can be accessed via 2 methods; by resolving locations using Location Types in conjunction with the Location Master database (proprietary TWC internally used geocoding services) or by supplying latitude and longitude coordinates as described below. The Weather APIs use the ISO 3166 standard for Country Codes. REF - ISO Standard Online Browsing Platform (OSB): https://www.iso.org/obp/ui/#search/code/

TWC Platforms and Geographic Coverage

Weather Channel API's are available for use against vast expanses of geographical areas and in many cases worldwide. To determine if an API you are interested in can be used for a specific area of the globe, please consult this link to the Country and Platform Coverage Cross Reference

Supported Country Postal Codes

- United States
- Country Code: US

- United Kingdom
- Country Code: GB

- France
- Country Code: FR

- Germany
- Country Code: DE

- Italy
- Country Code: IT

Header Information

The following HTTP headers should be set to the appropriate values: * reference documentation - www.w3.org source documentation reference

Header	Documentation	Usage
Accept-Encoding: **gzip**	http://www.w3.org/Protocols/rfc2616/rfc2616-sec14.html#sec14.3	**REQUIRED**: All requests to the API's should request that the response be compressed. The only supported encoding format is: gzip
TWC-Date: *Tue, 15 Nov 1994 08:12:31 GMT*	Similar to: http://www.w3.org/Protocols/rfc2616/rfc2616-sec14.html#sec14.18	Only required for requests authenticating with a secret. See how to calculate the token below in the Appendix for details on the TWC-Date.
Authorization: TWC *{user token}*	http://www.w3.org/Protocols/rfc2616/rfc2616-sec14.html#sec14.8	Only required for requests authenticating with a secret. See how to calculate the token below in the Appendix.

Cross-Origin Resource Sharing

All TWC APIs support Cross-Origin Resource Sharing or CORS. For additional information on this communications mechanism, please refer to the following documents:

Resource	Documentation
W3 CORS Specification	http://www.w3.org/TR/cors/
CORS Tutorial	http://www.html5rocks.com/en/tutorials/cors/

URL Construction

The URLs are assembled logically so they can be easily readable and recognized by developers and API users to identify the content being provided and the filters (parameters) available.   The following table defines the components of the URL.

URL Explanation:
Example: http//api.weather.com/<version>/<location group>/<location>/<product group>/<product subgroup>.<file type>?apiKey={key value}&<.... additional *parameter*s>

Attribute	Description
hostname	hosted url path (api.weather.com)
version	current iteration (ex: "v1")
product group	product (ex: "observations")
product subgroup	sub product (ex: historical) - optional group
file type	"json" or "xml"
location group	API family group (ex: "location" or "geocode") - determines the type location type to reference

location	valid value for the Location_id**, (examples: 30339:4:US (postal code); 22.56/-84.04)

Location Type	Description	Example
Geocode	A combination of latitude and longitude. Uses WGS84 geocode coordinate reference system. https://www.w3.org/2003/01/geo/	.../v1/geocode/33.56/-84.04/…
County Code	A proprietary TWC county code supplied to customers via an external look up service.	.../v1/county/{countrycode}/{countycode}/...
State Code	A proprietary TWC state and country code supplied to customers via an external look up service.	.../v1/state/{countrycode}/{statecode}/…
Country Code	The Weather APIs use the ISO 3166 standard for Country Codes.	.../v1/country/{countrycode}/…
Postal Code	The Weather APIs support Postal Codes for the supported Countries. Postal Codes support is dependent on a proprietary Location Type and can be called as: **<Postal Code>:<Location Type 4>:<Country Code>**	.../v1/location/30075:4:US/….
Airport ID	The Weather APIs support Airport ID for commercial airports. Airport ID support is dependent on a proprietary Location Type and can be called as: **<Airport ID>:<Location Type 9>:<Country Code>**	.../v1/location/ATL:9:US/...
Zone ID	A named geographical zone.	.../v1/zone/{zoneid}/…

Global Query Parameters
The following represent global query parameters for use by most APIs requests.

Parameter	Name	Description	Example
Geocode	**geocode**	TWC uses valid latitudes and longitude coordinates to identify locations worldwide. Entering a latitude and longitude coordinate will return the closest reporting location (if available). Any API that accepts latitude and longitude as input will support the use of both periods or commas as decimal separators. Furthermore, within a single call no API will accept a mix of separators both values should use all commas or all periods but not both.	../v1/**geocode/34.063/-84.217/** **33.40/83.19** is the manner in which a latitude and longitude is normally written in the US.
API Key	**apiKey**	The “apiKey” is used for access using the api; usually customer specific.	../v1/test/30075:4:US/forecast.json?**apiKey=yourApiKey**

General Query Parameters
The following represent general query parameters for specific API requests. These query paramendf
ters do not apply to all APIs.
Each API User document will specify if any of the following general query parameters are available for the API:

Parameter	Name	Description	Example
Dates	startDate	Specific dates requests - used for API requesting past or historical data requests. Dates are requested using date format “YYYYMMDD”.	../v1/test/30075:4:US/<product>/<product subtype>.json?**startDate=20140215**

Date Ranges	startDate & endDate	Dates range requests - used for API requesting past or historical data requests. Dates are requested using date format “YYYYMMDD”.	../v1/test/30075:4:US/<product>/<productsubtype>.json?**startDate=20140215&endDate=20140301**
Days	days	Days parameter is used for incrementation of data usually in conjunction with dates or using current system date.	../v1/test/30075:4:US/<product>/<product subtype>.json?**days=2**
Hours	hours	Hours parameter is used for incrementation of data usually in conjunction with dates or using current system date.	../v1/location/30075:4:US/<product>/<product subtype>.json?**hours=2**
Language	language	The ”language” parameter drives the language of the response.	../v1/test/30075:4:US/forecast.json?**language=es-MX**
Unit of Measure	units	The unit of measure for the response. The following values are supported: • e = English units • m = Metric units • h = Hybrid units (UK) • s = Metric SI units (not available for all APIs) The TWC API supports English, Metric, and UK-Hybrid units of measure (UOM). Units of measure is not a required parameter for the url. In situations where the units of measure is supplied but no value is given, the API will not return an error but will default the unit of measure in the response to those specified by the language code. See the Language Codes section in the Appendix for a complete listing of all supported language codes and their default units of measure. Units of Measure can be overwritten by supplying the “units” parameter.	../v1/location/30075:4:US/forecast.json?**units=e** ...v1/service?**units=e** ...v1/service?**units=m** ...v1/service?**units=h** ...v1/service?**units=s**

Metadata / Echo Parameters

All TWC APIs contain a metadata header that will accompany the response back to the customer. The various field values within the metadata header are primarily used for diagnostic purposes by the TWC internal team. However, within the metadata there are also fields which can be leveraged by the consumer for similar purposes. These fields are known as ‘echo parameters’ and represent query arguments embedded within the API URL prior to the call. Within the metadata header section, the global parameters (i.e., latitude, longitude, location, language and units) are returned depending on the manner in which the API is called.

Note: Information pertaining to the fields of individual APIs will be handled in API User Documents and Content Usage Guides.

API Metadata	Description	Type	Range	Null	Sample	API Usages
version	the version number of the api	string		N	1.1	All SUN APIs
transaction_id	transaction id for this call of the api	integer		N	1407766348658:1285362530	All SUN APIs
expire_time_gmt	Data expiration time in unix seconds This will be the minimal expire time	epoch		N	1380170732	All SUN APIs
location_id	Echo parameter displaying the requested or postal code. This field is only populated when the API is called for a postal code	string	Any valid postal code	Y	30339	All SUN APIs
latitude	Echo parameter displaying the requested latitude for a location's forecast. This field is only populated when the API is called for a geocode	float	Any valid latitude value	Y	33.41	All SUN APIs
longitude	Echo parameter displaying the requested latitude for a location’s forecast. This field is only populated when the API is called for a geocode	float	Any valid longitude value	Y	-83.42	All SUN APIs .

language	Echo parameter displaying the default or requested translation language for response text.	string	Any language supported by TWC	N	en-US	All SUN APIs
units	Echo parameter displaying the default or requested units of measure (UOM) for various numeric values.	string	‘e’ or ‘m’ or ‘h’ or ‘s’	Y	e	All SUN APIs * except for Alerts
countryCode	Echo parameter displaying the country code. Only populated when the API is called for a country code	string	Any valid TWC country codes		US	
stateCode	Echo parameter displaying the state code. Only populated when the API is called for a state code	string	Any valid TWC state codes	Y	GA	
areaID	Echo parameter displaying the area ID Only populated when the API is called for a state code	string	Any valid TWC area ID.			

Error Handling

The response on error is always the same, allowing multiple error types to be returned within one response:

JavaScript Object Notation (JSON) Response	Extensible Markup Language (XML) Response

{
"metadata": {
"transaction_id": "1459351742810:-2012966722",
"status_code": 400
},
"success": false,
"errors": [
{
"error": {
"code": "NDF-0001",
"message": "There was no data found for your <product> query."
}
}
]
}
	<errorResponse>
	<success>false</success>
	<metadata>
	<transaction_id>1459352136214:2098414588</transaction_id>
	<status_code>400</status_code>
	</metadata>
	<errors>
	<error code="NDF-0001">
	<message>
	There was no data found for your <product> query.
	</message>
	</error>
	</errors>
	</errorResponse>

Status Code	Description
200	OK. The request has succeeded.
400	Bad request. The request could not be understood by the server due to malformed syntax or there is no data found for the location requested.
401	Unauthorized. The request requires authentication.
403	Forbidden. The server understood the request but is refusing to fulfill it.
404	Not found. The endpoint requested is not found.
500	Internal server error. The server encountered an unexpected condition which prevented it from fulfilling the request.

Handling Null Data Fields in API Response

If a data field is null then TWC APIs will return the appropriate field tags with the word “null”. There should be no occasion in which field tags do not appear in the response of an API due to lack of data.

JavaScript Object Notation (JSON) Response	**Extensible Markup Language (XML) Response**

"astronomy": [
{
"moon": {
"phase_desc": "Waning Gibbous",
"phase_icon": 19,
"local_date_moonset": "8/15/2014",
"local_time_moonset": "",
"local_date_moonrise": "8/15/2014",
"local_time_moonrise": "",
"moonset": "",
"moonrise": ""
},
"sun": {
"local_date_sunset": "8/15/2014",
"local_time_sunset": **null**,
"local_date_sunrise": "8/15/2014",
"local_time_sunrise": **null**,
"sunset": **null**,
"sunrise": **null**
	<astronomy>
	<moon>
	<phase_desc>Waning Gibbous</phase_desc>
	<phase_icon> 19</phase_icon>
	<local_date_moonset>"8/15/2014"</local_date_moonset>
	<local_time_moonset>**null**</local_time_moonset>
	<local_date_moonrise>”8/15/2014”</local_date_moonrise>
	<local_time_moonrise>**null**</local_time_moonrise>
	<moonset>**null**</moonset>
	<moonrise>**null**</moonrise>
	</moon>
	<sun>
	<local_date_sunset>"8/15/2014"</local_date_sunset>
	<local_time_sunset>**null**</local_time_sunset>
	<local_date_sunrise>"8/15/2014"</local_date_sunrise>
	<local_time_sunrise>**null**</local_time_sunrise>
	<sunset>**null**</sunset>
	<sunrise>**null**</sunrise>
	</sun>

API logic to handle forecast request data outages

Provide a method to lengthen API cache times for forecast data during any forecast generation outages.

Definitions	**Processing**	**Request made**	**Set Customer cache headers**
• mttl = maximum time to live (predefined at 12 hours) • ottl = optimum time to live (defined by FOD) • cb = circuit breaker - a manual set flag that indicates that FOD is down and the API should not try to fetch data from it.	• Set redis cache based on mttl	• ottl not expired, serve • ottl expired, cb green, fod fails then serve from cache • ottl expired, cb green, fod success, serve, update cache • ottl expired, cb red, serve from cache	• When otto is past we should set customer cache header to time + 15 • When otto is current we should set customer cache header to otto

Appendix

How to Calculate the Authorization Header

For API keys that use a shared secret, every HTTP request needs to calculate a token, and put that into a header.

Description	**Example**
First you must have your shared secret.	For this example, we’ll use: 12345

Second, you must set a HTTP header named **TWC-Date** which contains the current time (see clock synchronization and why TWC-Date). This is specified by RFC1123. This header must be in GMT, and it must be specified with the standard C language localization.	An example date would be: Fri, 10 Jan 2014 15:26:17 GMT
Third, you must calculate a portion of the URL. This is defined as "from the first slash, up to but not including the first question mark".	For example. If you are trying to retrieve: http://api.weather.com**/v2/forecast/daily/3day**?apiKey=<your_api_key> The url portion would be: /v2/forecast/daily/3day
Now you have all the pieces. Here's the **formula to calculate the token**: SHA256(<secret> + <date> + <url_portion>)	So, our example would be: SHA256("12345Fri, 10 Jan 2014 15:26:17 GMT/v2/forecast/daily/3day") which yields: 7dc9de03043885025e95889e960ca735ae09cfae3f47bf2393a46de8b7a2417c
Now you can put it all together. You make your HTTP GET request for: http://api.sun.weather.com/v2/forecast/daily/3day?apiKey=<your_api_key>	With two different request headers (again, see why TWC-Date): **TWC-Date:** Fri, 10 Jan 2014 15:26:17 GMT **Authorization: TWC** 7dc9de03043885025e95889e960ca735ae09cfae3f47bf2393a46de8b7a2417c

TWC-Date Clock Synchronization

The TWC-Date header needs to be generated by a relatively accurate clock. We have taken measures to keep our systems aligned to highly accurate clocks. Clients must keep their clocks accurate as well. We do allow client clocks to vary by +/-5 minutes (300 seconds). However, requests will fail if the specified TWC-Date header falls outside of this 5 minute window.

TWC-Date Format

The HTTP 1.1 specification RFC2616 in section 3.3.1 specifies 3 allowable Date specifications. We require the first one, specified in RFC822 section 5.1 and updated in RFC1123 section 5.2.14. In addition, we also require that all dates be in the GMT or UTC time zone.

C based languages can use this format string for strftime() and equivalents: ● "%a, %d %b %Y %H:%M:%S GMT" Example date strings include: ● Sat, 11 Jan 2014 15:26:07 GMT ● Fri, 04 Jul 2014 17:06:52 GMT ● Mon, 07 Jul 2014 09:59:21 GMT	**Why TWC-Date and not Date?** Some REST client libraries do not allow you to set the **Date** header. To facilitate this, we have moved to leveraging the **TWC-Date** header instead. The internal logic will always leverage the **TWC-Date** header, if present. The logic will fall back to using the **Date** header if **TWC-Date** is not present.

Units of Measure

Standard Units Of Measure (UOM) Description

Short Name	Long Name	Imperial (English) - e	Metric - m	Metric SI - s	Hybrid UK - h

temp	Temperature	f (fahrenheit)	c (celsius)	c (celsius)	c (celsius)
pressure	Pressure	hg (inches of mercury)	mb (millibars)	mb (millibars)	mb (millibars)
precip	Precip Amount	in (inches) - rain/snow	mm (millimeters - rain), cm (centimeters - snow)	mm (millimeters - rain), cm (centimeters - snow)	mm (millimeters - rain), cm (centimeters - snow)
distance	Distance	mi (miles)	km (kilometer)	m (meter)	mi (miles)
vis	Visibility	mi (miles)	km (kilometer)	km (kilometer)	km (kilometer)
wnd_speed	Wind Speed	mh (miles/hour)	km (kilometer/hour)	m/s (meters per second)	mph (miles per hour)
wave_height	Wave Height	ft (feet)	mtr (meter)		ft (feet)

Unit of Measure Conversion Tables

Type	**Imperial to Metric**	**Metric to Imperial**
Temperature	Fahrenheit to Celsius: Divide (F - 32) by 1.8	Celsius to Fahrenheit: Multiply C by 1.8 and Add 32
Pressure	Inches of Mercury to Millibars: Multiply Inches by 33.8637526	Millibars to Inches of Mercury : Multiply Millibars by 0.0295301
Distance/Speed	**Imperial to Metric**	**Metric to Imperial**
	Miles to Kilometers: Multiply Miles by 1.60934	Kilometers to Miles : Multiply Kilometers by 0.6214
	Feet to Meters: Multiply Feet By 0.3048	Meters to Feet: Multiply Meters By 3.2808
	Miles per Hour (MPH) to Meters Per Second: Multiply MPH by 0.44704	Meters Per Second(M/S) to Miles per Hour: Multiply M/S by 2.2369362920544
	Inches to Centimeters: Multiply Inches by 2.54	Centimeters to Inches: Multiply Centimeters by 0.39370
	Inches to Millimeters: Multiply Inches by 25.4	Millimeters to Inches: Multiply Centimeters by 0.0393701
		Metric Conversions
		Kilometer per Hour (KPH) to Meters Per Second: Multiply KPH by 0.277778 Meters Per Second (M/S)to Kilometer per Hour: Multiply M/S by 3.6

Visibility Standards

For API products that report visibility

Field	Original Value	New Value

Summary	Visibility should have 3 digits of precision for English units	Visibility should have 3 (English) or 2 (Metric) digits of precision for less 1 else Integer
Description	The visibility value in English units of measurement should be 3 digits of precision.	The rule should be: > 1 to 999 should be an integer (greater than 1 = no decimal) For less than 1 = 2 (metric) & 3 (English) decimal places

Language Codes

	Language Code	Description	Short Language Code	Unit of Measure
1	ar-AE	Arabic - UAE	ar	m
2	bn-BD	Bengali (Bangladesh)	bn	m
3	bn-IN	Bengali (India)		m
4	ca-ES	Catalan	ca	m
5	cs-CZ	Czech (Czech Republic)	cs	m
6	da-DK	Danish	da	m
7	de-DE, de-CH	German/Germany, German/Switzerland	de	m
8	el-GR	Greek	el	m
9	en-GB, en-AU	English/Great Britain, English/Australia		h
10	en-IN	English/India		m
11	en-US	English/United States	en	e
12	es-AR	Spanish/Argentina		m
13	es-ES, es-US	Spanish/Spain,Spanish/US		m
14	es-LA	Spanish (LATAM)		m
15	es-MX	Spanish/Mexico		m
16	es-UN	Spanish (International)		m
17	es-ES	Spanish/Spain	es	e
18	fa-IR	Persian (Farsi)	fa	m
19	fi-FI	Finnish	fi	m
20	fr-CA	French (Canada)		m
21	fr-FR, fr-CH	French/France, French/Switzerland	fr	m
22	he-IL	Hebrew	he	m
23	hi-IN	Hindi	hi	m
24	hr-HR	Croatian	hr	m
25	hu-HU	Hungarian	hu	m
26	in-ID	Indonesian	in	m
27	it-IT, it-CH	Italian/Italy, Italian/Switzerland	it	m
28	iw-IL, he-IL	Hebrew	iw	m
29	ja-JP	Japanese	ja	m
30	kk-KZ	Kazakh	kk	m

31	ko-KR	Korean	ko	m
32	ms-MY	Malay	ms	m
33	nl-NL	Dutch	nl	m
34	no-NO, nn-NO, nn	Norwegian	no	m
35	pl-PL	Polish	pl	m
36	pt-BR	Portuguese/Brazil	pt	m
37	pt-PT	Portuguese		m
38	ro-RO	Romanian	ro	m
39	ru-RU	Russian	ru	m
40	sk-SK	Slovak - Slovakia	sk	m
41	sv-SE	Swedish - Sweden	sv	m
42	th-TH	Thai - Thailand	th	m
43	tl-PH	Tagalog (Philippine)	tl	m
44	tr-TR	Turkish - Turkey	tr	m
45	uk-UA	Ukrainian - Ukraine	uk	m
46	ur-PK	Urdu	ur	m
47	vi-VN	Vietnamese	vi	m
48	zh-CN	Simplified Chinese		m
49	zh-HK	Chinese/Hong Kong	zh	m
50	zh-TW	Chinese/Taiwan		m